Fejér's Theorem

An End-to-End Meta-Math Walkthrough

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Abstract

A complete authored walkthrough of fejér's theorem, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

HA-02 / Harmonic Analysis / Report v1

I. TARGET AND IMPORT CONTRACT

Theorem I.1. *If f is continuous and 2π-periodic, the Cesàro means of its Fourier partial sums converge uniformly to f:*

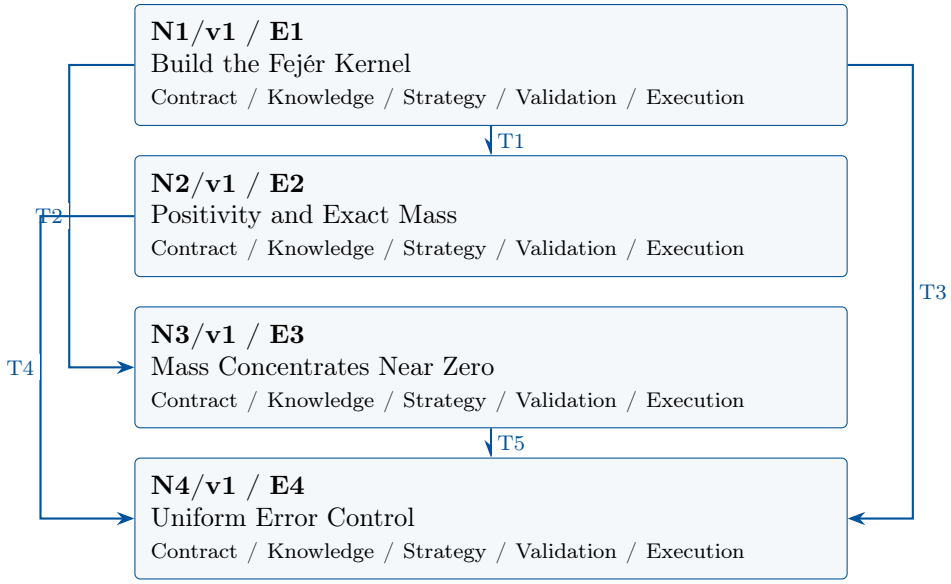
$$\sigma_N f = \frac{S_0 f + \cdots + S_N f}{N+1}, \qquad \|\sigma_N f - f\|_\infty \to 0.$$

Use normalized measure $dt/(2\pi)$ on $[-\pi, \pi]$, coefficients $c_k = \int f(t)e^{-ikt}\, dt/(2\pi)$, and $S_n f(x) = \sum_{|k|\le n} c_k e^{ikx}$. The output concerns averaged partial sums, not a promise about the raw S_n.

A. Foundations and permitted imports

elementary exponential orthogonality, finite geometric sums, boundedness and uniform continuity of continuous functions on a compact circle. All kernel identities and convergence estimates are proved below.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. $\sigma_N f = f * K_N$ and $K_N(t) = \frac{1}{N+1}|\sum_{j=0}^{N} e^{ijt}|^2$.

Receiving activity. N2: Positivity and Exact Mass

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N1/E1 and finite exponential orthogonality.

T2: N1/E1/v1 to N3. $\sigma_N f = f * K_N$ and $K_N(t) = \frac{1}{N+1}|\sum_{j=0}^{N} e^{ijt}|^2$.

Receiving activity. N3: Mass Concentrates Near Zero

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N1/E1, with a fixed distance δ from zero on the circle.

T3: N1/E1/v1 to N4. $\sigma_N f = f * K_N$ and $K_N(t) = \frac{1}{N+1}|\sum_{j=0}^{N} e^{ijt}|^2$.

Receiving activity. N4: Uniform Error Control

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N1/E1, N2/E2, N3/E3, and uniform continuity on the compact circle.

T4: N2/E2/v1 to N4. $K_N \geq 0$ and $\int_{-\pi}^{\pi} K_N(t)\, dt/(2\pi) = 1$.

Receiving activity. N4: Uniform Error Control

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N1/E1, N2/E2, N3/E3, and uniform continuity on the compact circle.

T5: N3/E3/v1 to N4. $\int_{\delta \leq |t| \leq \pi} K_N(t)\, dt/(2\pi) \leq [(N+1)\sin^2(\delta/2)]^{-1}$ for $0 < \delta < \pi$.

Receiving activity. N4: Uniform Error Control

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N1/E1, N2/E2, N3/E3, and uniform continuity on the compact circle.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Build the Fejér Kernel

Object and role. Build the Fejér Kernel. Establish the following local result: $\sigma_N f = f * K_N$ and $K_N(t) = \frac{1}{N+1}|\sum_{j=0}^{N} e^{ijt}|^2$.

Inputs. Normalized circle measure, Fourier coefficients, and finite geometric sums.

C — Contract

Own E1/v1: $\sigma_N f = f * K_N$ and $K_N(t) = \frac{1}{N+1}|\sum_{j=0}^{N} e^{ijt}|^2$.

K — Knowledge

Fourier conventions

S — Strategy

Rewrite average as convolution

V — Validation

Track N+1 and normalized measure

E — Execution

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 Write each partial sum as convolution with its Dirichlet kernel.

N1.2 Average the finite sums to obtain triangular Fourier weights.

N1.3 Expand the squared finite exponential sum and derive the sine quotient, with its continuous value at zero.

Export E1/v1. $\sigma_N f = f * K_N$ and $K_N(t) = \frac{1}{N+1}|\sum_{j=0}^{N} e^{ijt}|^2$.

Acceptance gate. Track N+1 and normalized measure Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: Track N+1 and normalized measure Recheck dependent nodes N2, N3, N4.

B. N2/v1: Positivity and Exact Mass

Object and role. Positivity and Exact Mass. Establish the following local result: $K_N \geq 0$ and $\int_{-\pi}^{\pi} K_N(t)\, dt/(2\pi) = 1$.

Inputs. N1/E1 and finite exponential orthogonality.

C — Contract

Own E2/v1: $K_N \geq 0$ and $\int_{-\pi}^{\pi} K_N(t)\, dt/(2\pi) = 1$.

K — Knowledge

Orthogonality

S — Strategy

Use squared finite sum

V — Validation

Include value at zero

E — Execution

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 Use the squared-sum identity for nonnegativity.

N2.2 Integrate the finite Fourier expansion: only its constant coefficient remains.

N2.3 Deduce preservation of constants and the sup-norm contraction bound.

Export E2/v1. $K_N \geq 0$ and $\int_{-\pi}^{\pi} K_N(t)\, dt/(2\pi) = 1$.

Acceptance gate. Include value at zero Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: Include value at zero Recheck dependent nodes N4.

C. N3/v1: Mass Concentrates Near Zero

Object and role. Mass Concentrates Near Zero. Establish the following local result: $\int_{\delta\le|t|\le\pi} K_N(t)\,dt/(2\pi) \le [(N+1)\sin^2(\delta/2)]^{-1}$ for $0<\delta<\pi$.

Inputs. N1/E1, with a fixed distance δ from zero on the circle.

C **Contract**

Own E3/v1: $\int_{\delta\le|t|\le\pi} K_N(t)\,dt/(2\pi) \le [(N+1)\sin^2(\delta/2)]^{-1}$ for $0<\delta<\pi$.

K **Knowledge**

Sine denominator bounds

S **Strategy**

Bound exterior mass

V **Validation**

Stay away from zero modulo 2π

E **Execution**

Carry out the numbered deductions and publish E3/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N3.1 Bound the sine numerator by one.

N3.2 Bound the denominator below by $\sin^2(\delta/2)$.

N3.3 Integrate over a set of normalized measure at most one.

Export E3/v1. $\int_{\delta\le|t|\le\pi} K_N(t)\,dt/(2\pi) \le [(N+1)\sin^2(\delta/2)]^{-1}$ for $0<\delta<\pi$.

Acceptance gate. Stay away from zero modulo 2π Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N3: Stay away from zero modulo 2π Recheck dependent nodes N4.

D. N4/v1: Uniform Error Control

Object and role. Uniform Error Control. Establish the following local result: $\|\sigma_N f - f\|_\infty \to 0$ for every continuous periodic f.

Inputs. N1/E1, N2/E2, N3/E3, and uniform continuity on the compact circle.

C — Contract

Own E4/v1: $\|\sigma_N f - f\|_\infty \to 0$ for every continuous periodic f.

K — Knowledge

Uniform continuity and boundedness

S — Strategy

Split near and far errors

V — Validation

Use one delta for all x

E — Execution

Carry out the numbered deductions and publish E4/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N4.1 Write the error as the convolution of the increment $f(x-t) - f(x)$.

N4.2 Choose one δ controlling the near part uniformly in x.

N4.3 Bound the far part by $2\|f\|_\infty$ times the tail mass and then choose N.

Export E4/v1. $\|\sigma_N f - f\|_\infty \to 0$ for every continuous periodic f.

Acceptance gate. Use one delta for all x Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N4: Use one delta for all x Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Build the Fejér Kernel

The Dirichlet kernel is $D_n(t) = \sum_{|k|\le n} e^{ikt}$. Finite rearrangement gives

$$S_n f(x) = \int_{-\pi}^{\pi} f(x-t) D_n(t) \, \frac{dt}{2\pi}.$$

Averaging yields the same expression with

$$K_N(t) = \frac{1}{N+1} \sum_{n=0}^{N} D_n(t) = \sum_{|k|\le N} \left(1 - \frac{|k|}{N+1}\right) e^{ikt}.$$

To obtain positivity, expand the square:

$$\left| \sum_{j=0}^{N} e^{ijt} \right|^2 = \sum_{j,l=0}^{N} e^{i(j-l)t} = \sum_{|k|\le N} (N+1-|k|) e^{ikt}.$$

Thus

$$K_N(t) = \frac{1}{N+1} \left| \sum_{j=0}^{N} e^{ijt} \right|^2 = \frac{1}{N+1} \frac{\sin^2((N+1)t/2)}{\sin^2(t/2)}$$

for t not a multiple of 2π. The continuous value at zero is N+1. This proves $\sigma_N f = f * K_N$ without any interchange of infinite sums.

B. N2 — Positivity and Exact Mass

The squared-sum expression proves $K_N \ge 0$. Integrate its finite Fourier expansion. Every nonzero exponential has normalized integral zero, while the constant coefficient is 1. Hence

$$\int_{-\pi}^{\pi} K_N(t) \, \frac{dt}{2\pi} = 1.$$

Consequently convolution preserves constants and $\|\sigma_N f\|_\infty \le \|f\|_\infty$.

C. N3 — Mass Concentrates Near Zero

Fix $0 < \delta < \pi$. For $\delta \le |t| \le \pi$, $\sin^2(t/2) \ge \sin^2(\delta/2)$, and the numerator is at most 1. Therefore

$$K_N(t) \le \frac{1}{(N+1)\sin^2(\delta/2)},$$

and, since the normalized measure of the integration set is at most 1,

$$\int_{\delta\le|t|\le\pi} K_N(t)\,\frac{dt}{2\pi} \le \frac{1}{(N+1)\sin^2(\delta/2)} \to 0.$$

The height at zero grows, but the mass stays one and moves into smaller neighborhoods. Height alone is not the approximation argument.

D. N4 — Uniform Error Control

Write $M = \|f\|\infty$. For any $\epsilon > 0$, uniform continuity on the circle gives delta such that $|f(x-t) - f(x)| < \varepsilon/2$ whenever $|t| < \delta$, uniformly in x. By positivity and unit mass,

$$\begin{aligned}|\sigma_N f(x) - f(x)| &\le \int |f(x-t) - f(x)| K_N(t)\,\frac{dt}{2\pi} \\ &\le \frac{\varepsilon}{2} + \frac{2M}{(N+1)\sin^2(\delta/2)}.\end{aligned}$$

Choose N so the second term is smaller than epsilon/2. This choice is independent of x, proving uniform convergence. □

A reusable quantitative version is

$$\|\sigma_N f - f\|_\infty \le \omega_f(\delta) + \frac{2\|f\|_\infty}{(N+1)\sin^2(\delta/2)},$$

where $\omega_f(\delta) = \sup_{x,\,|t|\le\delta} |f(x-t) - f(x)|$ on the circle. Its two terms encode different obligations: local continuity and distant kernel mass.

□

V. WORKED ORCHESTRATION

Let $f(x) = |x|$ on $[-\pi, \pi]$, extended periodically. The endpoint values match, so this defines a continuous function on the circle. It has $M = pi$ and a circle modulus bounded by delta. The error bound becomes

$$\|\sigma_N f - f\|_\infty \le \delta + \frac{2\pi}{(N+1)\sin^2(\delta/2)}.$$

The coefficients follow by integration:

$$c_0 = \frac{\pi}{2}, \qquad c_k = \frac{(-1)^k - 1}{\pi k^2} \quad (k \ne 0).$$

Hence

$$\sigma_N f(x) = \frac{\pi}{2} + 2\sum_{k=1}^{N} \left(1 - \frac{k}{N+1}\right) \frac{(-1)^k - 1}{\pi k^2} \cos(kx).$$

For $N = 1$ this is $\pi/2 - (2/\pi)\cos x$. The sequence smooths the corner while converging uniformly to it. To guarantee error below 0.1, for example, choose $\delta = 0.05$ and any integer N satisfying $2\pi/((N+1)\sin^2(0.025)) < 0.05$. This conservative bound proves a tolerance rather than merely displaying a good plot.

A. Boundary and hypothesis audit

Discontinuous periodic inputs cannot be uniform limits of these continuous trigonometric polynomials. A discontinuity therefore requires a different convergence statement. For a constant function the approximation is exact for every N. For any finite frequency k, the coefficient multiplier tends to one but is not identically one at finite N.

Export the convolution identity, positive unit-mass kernel, tail bound, and uniform approximation theorem. Combined with density of continuous functions in L^2, these support the completeness argument in Parseval's theorem.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

A failed branch replaces σ_N by S_N while retaining the positive-kernel proof. Validation observes that the Dirichlet kernel changes sign, so the positivity estimate is unavailable.

- **Node:** N4
- **State:** STALLED
- **Failing component:** Validation
- **Failed hypothesis:** nonnegative averaging kernel
- **Observed:** raw partial sum uses Dirichlet kernel
- **Repair target:** N1
- **Blocked export:** uniform convergence under the Fejer proof

Knowledge retrieves the Cesàro definition. Validation distinguishes the two kernels. Strategy restores averaging. Execution recomputes the squared-sum identity, positivity, and mass. Contract exports the theorem for σ_N and records that convergence of raw S_N requires a separate argument with additional considerations.

B. Hypothesis boundary

Discontinuous periodic inputs cannot be uniform limits of these continuous trigonometric polynomials. A discontinuity therefore requires a different convergence statement. For a constant function the approximation is exact for every N. For any finite frequency k, the coefficient multiplier tends to one but is not identically one at finite N.

Export the convolution identity, positive unit-mass kernel, tail bound, and uniform approximation theorem. Combined with density of continuous functions in L^2, these support the completeness argument in Parseval's theorem.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. MIT Fejér theorem lecture notes.